



General Framework

0.1.1 Basic Settings

Let

$$\mathbf{x}_t \in \mathbb{R}^{n_x}$$

denote the *state* vector at time t , which represents the unknown quantity we want to infer.

At each time t , we have *observation*

$$\mathbf{z}_t \in \mathbb{R}^{n_z},$$

which contains information about the hidden state $\mathbf{x}_t$, but may be noisy and uncertain. For convenience, define the observation history up to time t as

$$\mathbf{z}_{1:t} = \{\mathbf{z}_1, \dots, \mathbf{z}_t\}$$

Similarly, suppose that a *control input* is available at each time t , denoted by

$$\mathbf{u}_t \in \mathbb{R}^{n_u},$$

with the corresponding control history

$$\mathbf{u}_{1:t} = \{\mathbf{u}_1, \dots, \mathbf{u}_t\}.$$

Therefore, the available information evolves sequentially as

$$\mathbf{u}_1, \mathbf{z}_1, \mathbf{u}_2, \mathbf{z}_2, \mathbf{u}_3, \mathbf{z}_3, \dots$$

whereas the states

$$\mathbf{x}_1, \mathbf{x}_2, \dots$$

are not directly observed.

Since subsequent inference depends on the state, the objective is to characterize the uncertainty associated with the random variable $\mathbf{x}_t$ given all available information. This uncertainty is represented by the *belief state*

$$\text{bel}(\mathbf{x}_t) = p(\mathbf{x}_t | \mathbf{z}_{1:t}, \mathbf{u}_{1:t}).$$

The belief state is therefore a probability distribution over the possible values of $\mathbf{x}_t$ conditioned on the available observation and control histories. It is also commonly referred to as the posterior state distribution, information state, or state of knowledge.



0.1.2 Core Formula of Bayesian Filter

The key idea of the Bayesian filter is to represent uncertainty explicitly using the calculus of probability theory.

At each time step, the hidden state $\mathbf{x}_t$ evolves from the previous state $\mathbf{x}_{t-1}$ under the control input $\mathbf{u}_t$, while the observation $\mathbf{z}_t$ is generated from the corresponding state $\mathbf{x}_t$, as shown in Figure 0.1.

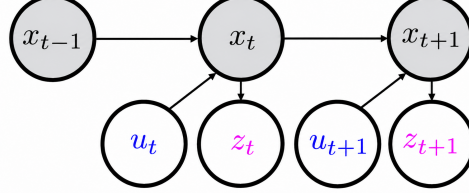


Figure 0.1 Graphical representation of the Bayesian filtering model

However, the belief, $p(\text{current state}|\text{all past information})$, is intractable. This leads to the Markov assumption: future state conditionally independent of past control inputs and observations given present state. We therefore have

$$p(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{u}_t, \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t-1}) = p(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{u}_t),$$

$$p(\mathbf{z}_t|\mathbf{x}_t, \mathbf{u}_t, \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t-1}) = p(\mathbf{z}_t|\mathbf{x}_t).$$

Using Bayes' rule and the Markov assumptions,

$$\begin{aligned}
\text{bel}(\mathbf{x}_t) &= p(\mathbf{x}_t|\mathbf{z}_{1:t}, \mathbf{u}_{1:t}) = p(\mathbf{x}_t|\mathbf{z}_t, \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t}) && \leftarrow \text{Definition of belief} \\
&= \frac{p(\mathbf{x}_t, \mathbf{z}_t|\mathbf{z}_{1:t-1}, \mathbf{u}_{1:t})}{p(\mathbf{z}_t|\mathbf{z}_{1:t-1}, \mathbf{u}_{1:t})} && \leftarrow \text{Condition probability} \\
&= \eta_t p(\mathbf{x}_t, \mathbf{z}_t|\mathbf{z}_{1:t-1}, \mathbf{u}_{1:t}) && \leftarrow \text{Normalization constant} \\
&= \eta_t p(\mathbf{z}_t|\mathbf{x}_t, \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t}) p(\mathbf{x}_t|\mathbf{z}_{1:t-1}, \mathbf{u}_{1:t}) && \leftarrow \text{Bayes rule} \\
&= \eta_t p(\mathbf{z}_t|\mathbf{x}_t) p(\mathbf{x}_t|\mathbf{z}_{1:t-1}, \mathbf{u}_{1:t}) && \leftarrow \text{Markov assumption} \\
&= \eta_t p(\mathbf{z}_t|\mathbf{x}_t) \int p(\mathbf{x}_t, \mathbf{x}_{t-1}|\mathbf{z}_{1:t-1}, \mathbf{u}_{1:t}) d\mathbf{x}_{t-1} && \leftarrow \text{Marginalization} \\
&= \eta_t p(\mathbf{z}_t|\mathbf{x}_t) \int p(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t}) p(\mathbf{x}_{t-1}|\mathbf{z}_{1:t-1}, \mathbf{u}_{1:t}) d\mathbf{x}_{t-1} && \leftarrow \text{Probability chain rule} \\
&= \eta_t p(\mathbf{z}_t|\mathbf{x}_t) \int p(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t}) p(\mathbf{x}_{t-1}|\mathbf{z}_{1:t-1}, \mathbf{u}_{1:t-1}) d\mathbf{x}_{t-1} && \leftarrow \text{Unnecessary condition} \\
&= \underbrace{\eta_t p(\mathbf{z}_t|\mathbf{x}_t)}_{\text{correction}} \underbrace{\int p(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{u}_t) \text{bel}(\mathbf{x}_{t-1}) d\mathbf{x}_{t-1}}_{\text{prediction}}
\end{aligned}$$

where η_t is a normalization constant,

$$\eta_t^{-1} = p(\mathbf{z}_t|\mathbf{z}_{1:t-1}, \mathbf{u}_{1:t}) = \int p(\mathbf{z}_t|\mathbf{x}_t) p(\mathbf{x}_t|\mathbf{z}_{1:t-1}, \mathbf{u}_{1:t}) d\mathbf{x}_t$$



and $\text{bel}(x_0) = p(\mathbf{x}_0)$.

The Bayesian filter therefore provides a recursive update approach to calculate belief tractably consisting of two steps: prediction, which propagates the previous belief through the state transition model, and correction, which incorporates the latest observation through the observation model. Algorithm 1 gives the pseudocode of Bayesian filter.

Algorithm 1 Bayesian Filter

Require: Initial belief $\text{bel}(\mathbf{x}_0) = p(\mathbf{x}_0)$, controls $\mathbf{u}_{1:T}$, observations $\mathbf{z}_{1:T}$, transition model $p(\mathbf{x}_t \mid \mathbf{x}_{t-1}, \mathbf{u}_t)$, and observation model $p(\mathbf{z}_t \mid \mathbf{x}_t)$

- 1: **for** $t = 1, 2, \dots, T$ **do**
 - 2: $\overline{\text{bel}}(\mathbf{x}_t) \leftarrow \int p(\mathbf{x}_t \mid \mathbf{x}_{t-1}, \mathbf{u}_t) \text{bel}(\mathbf{x}_{t-1}) d\mathbf{x}_{t-1}$ ▷ Prediction
 - 3: $\eta_t^{-1} \leftarrow \int p(\mathbf{z}_t \mid \mathbf{x}_t) \overline{\text{bel}}(\mathbf{x}_t) d\mathbf{x}_t$ ▷ Normalization
 - 4: $\text{bel}(\mathbf{x}_t) \leftarrow \eta_t p(\mathbf{z}_t \mid \mathbf{x}_t) \overline{\text{bel}}(\mathbf{x}_t)$ ▷ Correction
 - 5: **end for**
 - 6: **return** $\text{bel}(\mathbf{x}_T)$
-